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Effect of process-based strategies on anaerobic digestion of 3D-printing--derived polylactic acid: biogas production and mechanistic interpretation

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ABSTRACT

The rapid growth of 3D printing technology has led to an increase in polymeric waste, which requires proper management to avoid further pressure on the end-of-waste system. In this context, anaerobic digestion (AD) is a promising strategy for biorenewable energy production (e.g., biomethane) when biodegradable polymers, such as polylactic acid (PLA), are employed. Indeed, the latter has been widely used as a bioactive polymer for 3D printing, but the high printing temperatures can alter its properties, thereby affecting its biodegradability. This study investigated, for the first time, the biodegradation of 3D-printing PLA waste via AD under mesophilic conditions for 63 days. With the aim of improving the biodegradability of 3D-printing-derived waste, various process-based strategies, including micro-aeration, Fenton oxidation, and thermo-alkaline pretreatment, were thus employed. Biogas production and speciation during AD, as well as weight loss of PLA samples (i.e., untreated, unprinted, and printed), and their further characterization through differential scanning calorimetry analysis, FT-IR spectroscopy, and viscometrical measurements were assessed to study the effectiveness of the investigated process-based strategy. The combined thermo-alkaline pretreatment (i.e., 2.5 $\%_{w/v}$ of $Ca(OH)_2$ at 70 °C for 72 h) applied on 3D-printed PLA samples resulted in a total weight loss of 83.1% after AD process and a biomethane yield of approximately 360 mL/gPLA, which were significantly higher than that achieved with the control samples (i.e., 30% and <100 mL/gPLA, respectively). These results provided practical insights for the integration of additive manufacturing waste into AD-based circular economy strategies.

1. Introduction

Three-dimensional (3D) printing is a manufacturing process used to make objects from a digital Computer-Aided Design (CAD) model through a layer-by-layer material deposition approach. Among the various additive manufacturing technologies developed in the last decades, fused deposition modeling (FDM) is one of the most widely used open-source technologies, being FDM printers compact, affordable, and with low effort for minimal maintenance [1]. In general, during this process, a thermoplastic filament is heated above its melting point, extruded through a nozzle under controlled shear, and deposited in successive layers, where it quickly solidifies upon cooling to generate the final element. Although 3D printing was originally developed for rapid prototyping and modeling, it has evolved into a promising and widely adopted method for producing final, end-use products across multiple sectors, including consumer goods, biomedical devices, and engineering applications [2].

Despite the rapid development of FDM led to increased material throughput, it has also placed pressure on the waste management system due to a significant stream of polymeric waste. For instance, Song and

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Telenko [3] reported that polymeric waste from FDM printing accounts for 34% of the plastic used. Similarly, it has been reported that at least 5000 tons of waste were produced in 2020, over an annual consumption for 3D printing of 18,500 tons of plastic material [4].

Several approaches can be used to address the plastic waste issue management, including re-extrusion and the use of biodegradable plastics. About this, one of the main biodegradable plastics involved in 3D printing is indeed represented by polylactic acid (PLA) [5]. The latter is more environmentally friendly than conventional polymers (e.g., polyethylene), but 3D printing can affect this beneficial feature. Indeed, PLA filaments undergo repeated thermal and mechanical stresses during the FDM process, which can induce chain scission, reduce molecular weight, and modify crystallization behavior [6,7]. Therefore, due to the loss of mechanical properties after many recycling cycles, manufacturers are discouraged from adopting this practice. In addition, the FDM process could influence the biodegradability of FDM waste, casting doubt on the applicability of eco-friendly treatment approaches as alternatives to landfilling, such as biological processes.

Among biological processes, anaerobic digestion (AD) represents a suitable option to deal with biodegradable waste due to its effectiveness in decreasing the waste amount while producing biorenewable energy through biomethane production. Indeed, PLA is considered a biodegradable biopolymer according to international standards, such as EN 13432, ASTM D6400, ISO 14853, and ISO 17088, which define the requirements for aerobic/anaerobic biodegradation under controlled industrial composting/AD conditions [8]. For instance, it was observed that mesophilic AD carried out for over 75-90 days resulted in a 1-7% PLA degradation, while higher biodegradation (more than 80%) and biomethanation were achieved in about 2 months, but through thermophilic conditions [9–14]. However, due to the changes in the physical-chemical properties of FDM waste (i.e., surface areas, molecular weight, crystallinity, etc.) compared to the pristine materials [15], AD effectiveness can be altered.

Since thermophilic AD can be energetically unfeasible (i.e., negative energy balance) [16], the limited anaerobic conversion of PLA under mesophilic conditions represents a relevant technological challenge and one of the main motivations of the present study, which aimed at evaluating whether different process-based strategies could enhance the anaerobic digestion of 3D-printing-derived PLA waste and improve biogas production.

Therefore, to enhance the conversion of recalcitrant substrates, several process-based strategies, such as micro-aeration, Fenton oxidation, and thermo-alkaline pretreatment, have been developed to improve hydrolysis and biodegradation. Micro-aeration [17,18], a controlled oxygen dosing, was applied during AD to stimulate hydrolytic and facultative microbial activity, enhance redox balance, and accelerate the breakdown of recalcitrant waste. Also, oxidative pretreatments (e.g., Fenton-like processes) generate reactive oxygen species that can induce surface oxidation and partial chain cleavage [19,20]. Thermo-alkaline pretreatments, including $Ca(OH)_2$ addition at mild temperatures (e.g., 70 °C), can promote ester bond hydrolysis via nucleophilic attack, thereby reducing molecular weight and increasing solubility [21,22].

Although these approaches were mostly investigated for lignocellulosic materials (LMs), their potential impact on 3D-printed PLA waste, i.e., material already thermo-mechanically degraded [23,24], remains unexplored. For instance, a previous study reporting cold plasma treatment on PLA for food packaging also displayed an increase of the material hydrophilic nature and shorter molecular chains, both providing better conditions for microbial attachment on the treated PLA surface and fragmentation [25]. Also, previous studies only investigated the effect of other process-based strategies (i.e., PLA codigestion during AD) [26] or used pristine material in forms of pellets [27,28], sometimes mechanically reduced to powder [10,28,29], films [25,29], or pieces obtained from commercial items [30,31].

Therefore, to fill this knowledge gap, the present work focused on: (*i*) the investigation of AD for unprinted PLA (UnP-PLA) filament and printed PLA (P-PLA); (*ii*) the comparative evaluation of different process-based strategies (i.e., micro-aeration, Fenton-oxidation, and thermo-alkaline pretreatment) on the thermo-mechanically modified PLA; (*iii*) the mechanistic interpretation of the observed biodegradation behavior via integrated FT-IR, DSC, viscometry, weight-loss, and soluble chemical oxygen demand (COD) analyses; (*iv*) a kinetic evaluation based on the modified Gompertz model to investigate methane-production dynamics under the different process-based strategies and (*v*) the identification of the most effective strategy for enhancing biomethane production from 3D-printed PLA after a preliminary techno-economic assessment.

2. Materials and methods

2.1. Materials

2.1.1. Polylactic acid and printing procedure

A white PLA Ingeo 850 filament, purchased from Eolas Prints (Spain), was used for this study. This is a highly crystalline PLA with a diameter of 1.75 mm, a density of 124 g/cm^3, and a melting point of 165 – 180 °C. The UnP-PLA filament used for the experiments was cut into 3.8 cm-long rods weighing 0.109 ± 0.005 g. The P-PLA sticks were printed on a Delta WASP 2040 PRO 3D printer equipped with a SPITFIRE extruder at 195 °C, with the print bed maintained at 50 °C. Specifically, the P-PLA was vertically printed on a 0.5 cm-thick base, with a resulting stick length of 4 cm and a diameter equal to that of the starting filament (1.75 mm) (Fig. 1). The obtained P-PLA sticks weighed 0.109 ± 0.005 g, equal to that of the UnP-PLA filaments. Both UnP-PLA filaments and P-PLA sticks were stored in an airtight bag containing silica gel to maintain the polymer at low humidity.

2.1.2. Anaerobic inoculum

The anaerobic inoculum used for the AD tests was obtained from the first-stage mesophilic anaerobic digester of a full-scale wastewater treatment plant (WWTP) in Monopoli (Puglia, Italy). During sampling, the sludge was sieved to remove particles larger than 0.4 mm. To reduce residual biodegradable organic matter prior to the biochemical methane potential (BMP) assays and to accurately detect biogas produced solely from PLA biodegradation, the sludge was preliminarily incubated using a procedure adapted from Elbeshbishy et al. [32]. In brief, the 150 mL inoculum was placed in 0.25 L in PYREX glass bottles and pre-incubated at 35 ± 2 °C for 10 days under anaerobic conditions by flushing the headspace with argon gas. After pre-incubation, the inoculum showed a concentration of total solids (TS), volatile solids (VS), soluble COD,

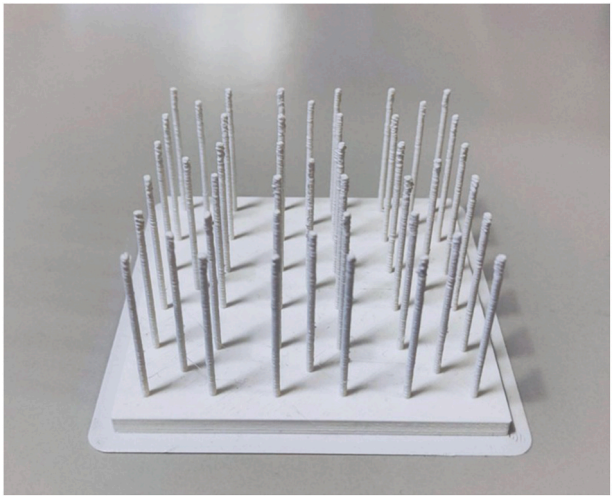


Fig. 1. P-PLA sticks produced from the 3D printing of PLA Ingeo 850 filament.

ammonia nitrogen (N–NH_4^+), and phosphate (P–PO_4^{3-}) of approximately 15.0, 10.50, 1.50, 0.49, and 0.04 g/L (Table 1), respectively.

Calcium hydroxide, sodium hydroxide, ferrous sulfate heptahydrate, hydrochloric acid (35%$_{v/v}$), and sulfuric acid (98%$_{v/v}$), used in the different pretreatments, were all purchased from Sigma Aldrich – Merck (Germany) and used as received.

2.2. Experimental procedures

2.2.1. Anaerobic digestion of UnP-PLA and P-PLA

The AD of PLA samples was carried out in PYREX glass bottles (0.25 L), closed with airtight caps equipped with a biogas sampling pipe, under mesophilic (35 ± 2 °C) conditions for 63 d, i.e., until the cumulative biogas production reached a plateau. The bottles were filled with 150 mL of inoculum and 0.65 g PLA as described in Table 2. Each bottle was flushed with Ar to remove air from the headspace. Under these conditions, considering a COD input of 1.5 gCOD/gPLA [28], the feed on microorganism ratio (F/M) was equal to 0.62 $gCOD_{PLA}/gVS_{inoculum}$, which can support high COD removal efficiencies [33]. Control tests (CTRL) were prepared under identical operating conditions, without PLA addition, to determine the contribution of PLA biodegradation to biogas production (mL). Each test was carried out in triplicate.

2.3. Process-based strategies for enhancing anaerobic biodegradation of PLA

2.3.1. Anaerobic digestion with micro-aeration approach

AD with micro-aeration (μAIR-AD) was also tested for 63 d on UnP-PLA and P-PLA. These tests were carried out in 2.5 L PYREX glass bottles, closed with airtight screw caps equipped with a biogas sampling pipe and a tube for the entry of air, under mesophilic (35 ± 2 °C) conditions. As described in Table 3, the reactors were filled with 1 L of inoculum, 4.16 g PLA, and 19 mL/d of air [34]. The air was insufflated throughout the whole experiment into the bottom of the reactors with a syringe pump to respect the optimal oxygen supply of 0.38 mL O_2/g VS/d [35]. Before the beginning of the AD tests, each reactor was flushed with Ar to remove air from the headspace. Control batch tests (CTRL), without PLA addition, were performed under the same operating conditions to distinguish the contribution of PLA biodegradation from endogenous sludge production. Each test was carried out in triplicate.

2.3.2. Anaerobic digestion with Fenton pretreatment

A ferrous solution was prepared by dissolving 388.8 mg $FeSO_4{\cdot}7H_2O$ in 200 mL aqueous solution adjusted at pH = 3 with HCl (35%$_{v/v}$). Afterwards, 0.65 g UnP-PLA or P-PLA were added to a beaker with 12.9 mL of ferrous solution. Then, 8 doses of 30 μL of H_2O_2 were added to each beaker every 5 min, and after 40 min, the pH was neutralized by adding NaOH. Finally, the soluble COD of the solution was measured, and the pretreated PLA sticks, namely UnP-PLA (F) and P-PLA (F), were washed, dried, weighed, and characterized.

A second set of PLA sticks, namely UnP-PLA (F + AD) and P-PLA (F + AD), also underwent the above-described Fenton pretreatment and were subsequently used in AD tests. For these tests, both the pretreated PLA sticks and the respective neutralized Fenton solutions were poured into 0.25 L PYREX glass bottles containing 0.15 L of inoculum and placed under mesophilic conditions (35 ± 2 °C) for 63 d. The reactors were hermetically sealed and equipped with a gas sampling port. Prior to incubation, the headspace was flushed with argon gas to ensure anaerobic conditions. Simultaneously, a blank test (CTRL) was also prepared without the addition of PLA. The tests were carried out in triplicate.

Table 2
Experimental design of AD without pretreatment.

Test	Inoculum (L)	UnP-PLA (g)	P-PLA (g)	F/M $\left(\frac{g\ COD_{PLA}}{g\ VS_{inoculum}}\right)$
CTRL	0.15	-	-	-
UnP-PLA (AD)	0.15	0.65 (±0.03)	-	0.62
P-PLA (AD)	0.15	-	0.65 (±0.03)	0.62

2.3.3. Anaerobic digestion with thermo-alkaline pretreatment

In this case, 0.65 g PLA samples were first pretreated with 13 mL alkaline solution (2.5 %$_{w/v}$ of $Ca(OH)_2$) at different temperatures (35 °C and 70 °C) and durations (24 h, 48 h, and 72 h). After the pretreatment, the pH was neutralized with HCl (35%$_{v/v}$) while the PLA sticks were dried and weighed, and the COD of the remaining solution was measured. Once the operating conditions providing the highest weight loss and solubilization of PLA samples were identified, the $Ca(OH)_2$ thermo-alkaline pretreatment was repeated on other UnP-PLA filaments and P-PLA sticks (samples labeled as UnP-PLA (AT) and P-PLA (AT), respectively). A part of these samples was collected for further properties characterization, while the remaining amount was used in AD treatments. In this latter case, the samples collected at the end of AD were labeled as UnP-PLA (AT + AD) and P-PLA (AT + AD). The biological reactors were set up as previously described for the Fenton-pretreated samples, and all experiments were carried out in triplicate.

2.4. Analytical methods

The content of TS and VS in the inoculum was measured by thermal processes, respectively at 105 °C and 550 °C, in agreement with APHA standard methods 2540-B and 2540-E [36].

All the liquid samples were centrifuged and filtered before the chemical analyses. The soluble COD of the samples was measured with the APHA standard method 5220-D [36] through an ONDA UV-30 SCAN UV-Visible spectrophotometer at 420 nm. The N-NH_4^+ concentration was measured using an ion chromatograph (Eco IC, Metrohm) equipped with a Metrosep C 4 - 250/4.0 column. The concentration of P-PO_4^{3-} was estimated through an Easychem Plus nutrient analyzer (Systea) following the US EPA method number 365.1.

The biogas volumes produced during the AD were periodically measured with an eudiometer and sampled to determine their composition in terms of CH_4 and CO_2. To this purpose, a gas chromatograph (8890 GC System, Agilent), equipped with a micro-packed column GC ShinCarbon ST, 80/100, 2 m, ID 0.53 mm (Restek) and a TCD detector, was used with Ar as gas carrier.

After pretreatments and digestion tests, PLA samples were washed with milliQ water and dried for 24 h at 40 °C in an oven. Then, the weight of the dried samples (W_{dry}) was measured. The weight loss of the samples was calculated as follows (Eq. (1)):

$$\textbf{\textit{Weight loss}} = \left(\frac{W_0 - W_{dry}}{W_{dry}}\right) \text{x } 100 \quad (1)$$

where W_0 is the sample weight before any pretreatment or AD test.

Afterwards, PLA sticks underwent a physical-chemical characterization to assess any changes in the structure of the PLA at the end of all the tested phases.

Differential scanning calorimetry (DSC) analyses were performed by a Mettler DSC 30 calorimeter with a heating-cooling-heating cycle in the

Table 1
Characteristics of mature inoculum.

	TS (g/L)	VS (g/L)	$COD_{soluble}$ (g/L)	N-NH_4^+ (mg/L)	P-PO_4^{3-}(mg/L)
Mature inoculum	15.0 ± 2.3	10.5 ± 1.7	1.5 ± 0.1	494 ± 50	40 ± 4

Table 3
Experimental design of AD with micro-aeration.

Test	Inoculum (L)	Airflow (mL/d)	UnP-PLA (g)	P-PLA (g)	F/M ($\frac{g\ COD_{PLA}}{g\ VS_{inoculum}}$)
CTRL	1	19	-	-	-
UnP-PLA (μAIR-AD)	1	19	4.16 (±0.11)	-	0.62
P-PLA (μAIR-AD)	1	19	-	4.16 (±0.11)	0.62

range 0-200 °C at rates of ±10 °C/min, flushing nitrogen at 100 ml/min on specimens of about 10 mg. Glass transition (T_g), crystallization (T_c), and melting (T_m) temperatures were evaluated. The degree of crystallinity (X_C) was assessed as (Eq. (2)):

$$X_C = \frac{\Delta H_m - \Delta H_C}{\Delta H_0}\ x\ 100 \quad (2)$$

where ΔH_0 is the heat of fusion of 100% crystalline PLA, taken equal to 93.6 J/g [37], and ΔH_c and ΔH_m are the experimental crystallization and melting enthalpy of the polymer, respectively. For further comparison, the crystallizability was also calculated in the second scan from the melting enthalpy (ΔH_{m2}) normalized to 93.6 J/g.

Molecular weight determinations were made by viscometric measurements on diluted chloroform/polymer solutions of about 0.3 g/dl in a Ubbelohde viscometer (type 0a) at 25.0 °C. The viscometric molecular weight, M_v, was calculated from the intrinsic viscosity, [η], using Eq. (3) [38]:

$$[\eta] = 5.45{\cdot}10^{-4}{\cdot}M_v^{0.73} \quad (3)$$

FT-IR spectra were acquired using a PerkinElmer spectrometer model. Spectrum One in attenuated total reflectance (ATR) mode in the wavenumber range of 4000 - 650 cm^{-1} with a resolution of 4 cm^{-1} and 4 scans was used.

All the FT-IR data were normalized for a straightforward comparison and semi-quantitative analysis. Transmission data were converted to absorbance and corrected with a piecewise-polynomial "rubber-band" baseline [39]. Anchor points were placed in the same absorption band-poor regions across the spectrum, including a hard constraint at 4000 cm^{-1}, and a low-order polynomial (3rd order) was fitted between anchors to remove slow background without distorting vibrational envelopes. This procedure is reported to be a useful and robust approach for baseline correction of vibrational spectra, which provides a stable basis for area normalization and quantification [40]. After baseline subtraction, each spectrum was normalized by the integrated area of the band at ~1454 cm^{-1} using a fixed window (1465–1440 cm^{-1}) in order to account for any bato- or ipsochromic shift of the absorption band across the samples. The ~1454 cm^{-1} feature in PLA is dominated by δ(CH_3) deformation, which is intense, relatively narrow, and comparatively insensitive to crystal-amorphous conversion [41,42]. Ordering of samples was quantified in the 980–900 cm^{-1} region by integrating the bands centered near 921 cm^{-1} and 956 cm^{-1}, both representing the coupled rocking vibration of -CH_3 moieties with skeletal stretching modes ν(C-C). These band intensities are environment-dependent: the ~921 cm^{-1} band reflects ordered 10_3 helical packing (α/α′ crystals), while ~956 cm^{-1} is enriched in the less-ordered (amorphous) configurations. Numerous previous studies [43–47] used these assignments from which an IR crystalline index can be derived as per Eq. (4):

$$\boldsymbol{IR - C_{index}} = \frac{A_{921}}{A_{921} + A_{956}}\ \text{x}\ 100 \quad (4)$$

which qualitatively tracks crystallinity features of PLA [48].

3. Results and discussion

3.1. Characterization of pristine PLA samples

Both UnP-PLA and P-PLA have been firstly characterized by means of DSC and viscometry, as reported in Table 4 and Table 5, respectively. DSC thermogram of pristine UnP-PLA (Fig. 2) in the first scan showed (i) a glass transition temperature at 66.4 °C immediately followed by an intense endothermic peak at 68 °C, due to the physical ageing; (ii) a crystallization peak in the range 90-135 °C (maximum at 117 °C); and (iii) a melting peak in the range of 135-170 °C (minimum at 153 °C). During the subsequent cooling, no crystallization phenomena were shown but only a T_g at 57.4 °C (Table S1 of the Supplementary information). Then, the second heating scan was typical of amorphous PLA, with T_g at 62 °C, crystallization peak at 119.0 °C (crystallization enthalpy of 22.6 J/g) and sharp melting peak at 152 °C (melting enthalpy of 23.7 J/g). The melting temperature of UnP-PLA was lower than that of enantiomerically pure poly-L-lactic acid (170-180 °C), indicating that about 5% of the pristine PLA was present in the D-form [49].

As reported in Table 4, on the first DSC scan, both pristine materials (UnP-PLA and P-PLA) were essentially amorphous ($X_C^{1st} = 2.0\%$ for UnP-PLA and 0.5% for P-PLA, calculated from Eq. (2)). The second heating likewise yielded uniformly low crystallinity ($X_C^{2nd} \approx$ 1.1-1.2%), confirming that crystallization was negligible during cooling step due to the relative high cooling rate of −10 °C/min (Fig. 2, Table S1 of the Supplementary information). In general, low values of crystallinity were observed in the second heating scan for all of the PLA samples,

Table 4
DSC results from the first heating scan of PLA samples. Glass transition temperature (T_g) and corresponding variation of specific heat (ΔC_p); crystallization temperature (T_c) and enthalpy (ΔH_c); melting temperature (T_m) and enthalpy (ΔH_m). Crystallinity (X_C) calculated from Eq. (2).

1st Heating scan							
Sample	T_g (°C)	ΔC_p (J/g•°C)	T_c (°C)	ΔH_c (J/g)	T_m (°C)	ΔH_m (J/g)	X_C (%)
UnP-PLA	66.4	0.50	116.9	23.4	152.7	25.3	2.0
P-PLA	64.9	0.55	116.5	26.8	153.3	27.3	0.5
UnP-PLA (AD)	68.7	0.59	106.9	25.6	149.7	28.6	3.2
P-PLA (AD)	67.2	0.61	105.7	29.3	156.3	29.8	0.5
UnP-PLA (μAIR-AD)	65.6	0.59	106.4	24.9	148.5	25.1	0.1
P-PLA (μAIR-AD)	65.9	0.52	107.2	25.7	149.0	27.9	2.4
P-PLA (F + AD)	64.3	0.54	106.2	24.0	148.7	27.1	3.3
UnP-PLA (F + AD)	64.1	0.56	105.7	24.7	148.5	29.6	5.2
UnP-PLA (AT)	63.0	0.30	-	-	156.7	41.8	44.7
P-PLA (AT)	61.3	0.30	-	-	155.3	46.8	50.0
UnP-PLA (AT + AD)	56.5	0.47	-	-	143.8	60.4	64.5
P-PLA (AT + AD)	52.8	0.44	-	-	144.2	49.2	52.6

Table 5
Intrinsic viscosity [η] and viscometric molecular weight M_v of various PLA samples (see details in the experimental part).

PLA Sample	[a]Intrinsic viscosity, [η] (dL/g)	[b]Viscosity-average molar weight, M_v (g/mol)
UnP-PLA	1.496	51300
P-PLA	1.125	34740
UnP-PLA (AD)	1.124	34690
P-PLA (AD)	1.056	31850
UnP-PLA (AT)	0.584	14140
P-PLA (AT)	0.431	9350

[a] Obtained by Huggins-Kraemer extrapolation to infinite dilution.
[b] Calculated from Eq. (3) [Schindler Harper].

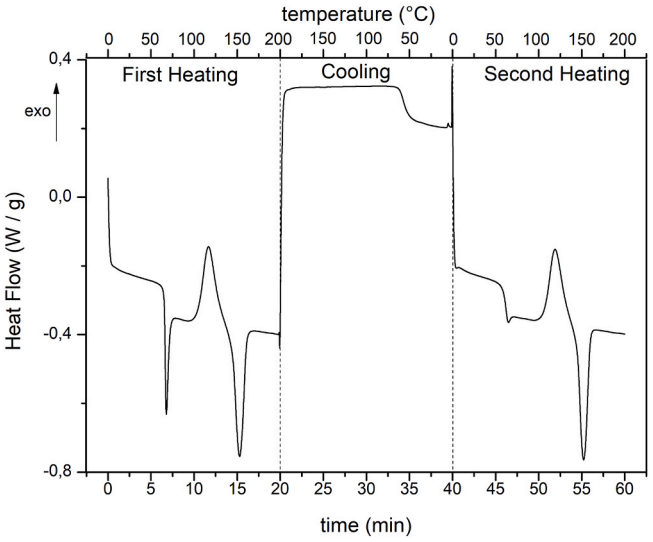


Fig. 2. DSC thermogram of pristine UnP-PLA: first heating scan, cooling and second heating scan.

indicating that the first melting deleted all previous thermal history.

Otherwise, the molecular weight of PLA after 3D printing decreased from 51300 to 34740 g/mol (Table 5) corresponding to an intrinsic viscosity decrease from 1.496 dL/g (UnP-PLA) to 1.125 dL/g (P-PLA). This can be attributed to thermo-mechanical chain-scission taking place during filament extrusion. Regarding the morphology of the samples, the images obtained through optical microscopy as reported in Fig. S1 of the Supplementary information, highlight a microscopic change of the P-PLA morphology resulting in ≈23% increase of the specific surface.

Although optical microscopy revealed a modest increase in specific surface area after printing, this morphological contribution was likely secondary to the reduction in molecular weight during PLA depolymerization and solubilization [50,51].

3.2. Effect of AD on PLA samples

Selected DSC thermograms of AD-treated PLAs are shown in Fig. 3, with evidence of glass transition temperature (T_g), crystallization peak (T_c) and melting peak (T_m) in the first heating scan. PLA after AD maintained an almost amorphous structure, as documented by the values of X_c reported in Table 4 (X_C^{1st} for UnP-PLA slightly increased to 3.2% while for P-PLA remained at 0.5%), similar to those of the pristine PLA.

Although thermal analyses do not directly quantify AD efficiency, changes in thermal transitions and degradation behavior can provide indirect evidence of structural modifications, depolymerization phenomena, and altered hydrolytic susceptibility occurring during the

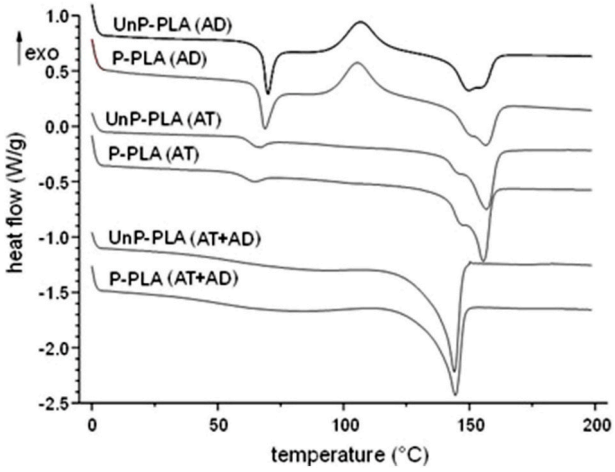


Fig. 3. Selected DSC thermograms: first heating scan of PLA samples (AD, AT and AT + AD).

biological process [52]. The second-run values were likewise small (UnP-PLA 0.7%, P-PLA 0.8%). A significant decrease of crystallization temperature in the first scan was also observed, from about 117 °C for pristine PLAs to about 107 °C, suggesting a reduction of molecular weight [49]. Similar behavior, even though to a lesser extent, was retrieved during the second scan (Table S1 of the Supplementary information). Only a slight decrease of crystallization peak from about 119 to 117-118 °C, and a slight increase of crystallizability from 25-28% to 28-32% were observed.

After the AD treatment, the UnP-PLA and P-PLA molecular weights respectively decreased to 34690 and 31850 g/mol (Table 5), thus confirming the previous hypothesis. However, the magnitude of these changes was modest, indicating that the observed chain shortening does not translate into measurable modifications in chain packing or crystalline fraction under these conditions. In addition, although a reduction in molar mass is generally expected to facilitate biodegradation [53,54], both P-PLA and UnP-PLA exhibited no mass loss after 63 days of mesophilic AD, and none of them measurably affected total biogas production (Fig. 4) or CO_2/CH_4 speciation compared to the CTRL test (data not shown). These outcomes align with numerous studies reporting limited degradability of PLA under mesophilic AD and indicate that neither 3D molding nor the slight molecular weight decrease that occurred during AD was sufficient to promote PLA degradation under these conditions [55–57].

In the IR overlays (Fig. S2 of the Supplementary information), the four spectra of UnP-PLA and P-PLA, before and after mesophilic AD, overlapped almost perfectly, especially in the relevant frequency window between 1900 and 650 cm^{-1}. The carbonyl region presented a single and intense ν(C=O) band centered at ~1747 cm^{-1} in all cases,

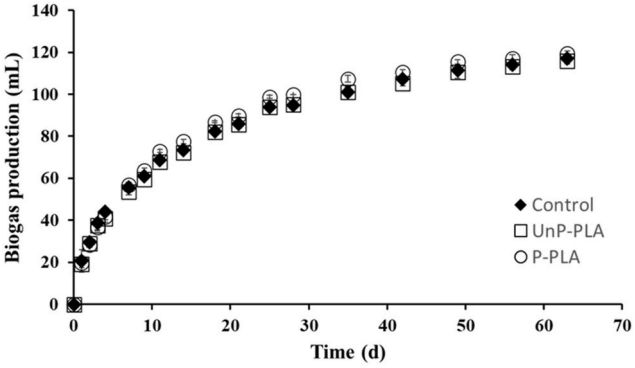


Fig. 4. Biogas production during the AD of UnP-PLA and P-PLA.

with no significant shifts across the samples, while the mixed C-O-C/C-C-O/CH_3-coupled manifold at ~1210-1040 cm^{-1} retained the same overall profile in frequency maxima and intensities [58–60]. Only in the crystallinity-sensitive range at 980–900 cm^{-1}, a small variation occurred, appreciable only when calculating the areas of the peaks: a slight attenuation near ~956 cm^{-1} accompanied by a slight growth at ~921 cm^{-1} (Table S2 of the Supplementary information). Accordingly, the ordering index (IR-C_{Index}) was very low and shifted only marginally (upon 3D printing of UnP-PLA: from 3.5 to 3.9; for UnP-PLA after AD: from 3.5 to 6.8; for P-PLA after AD: from 3.9 to 4.2), reinforcing the idea that neither printing nor digestion introduced new functionalities or significant changes in packing of PLA polymeric chains. These findings closely reflect the DSC results.

3.3. Effect of micro-aeration during AD of PLA samples

The introduction of a small amount of oxygen during AD (μAIR-AD) has been shown to be an effective approach to enhance the typical low hydrolysis rate of LMs or slow-biodegradable substrates [34,61,62]. Since no significant results were obtained during the AD of PLA, micro-aeration (μAIR) was tested with the aim of improving the substrate hydrolysis rate and its subsequent biodegradation. However, both UnP-PLA and P-PLA did not show a significant weight loss at the end of an AD process lasting 60 d and carried out with the addition of 19 mL/d of air. On the contrary, the μAIR-AD led to an increase of the average biogas production of 48% compared to the conventional AD (Fig. 5). In fact, an average amount of 109.7 $mL_{biogas}/g_{VSinoculum}$ and 74 $mL_{biogas}/g_{VSinoculum}$ were obtained at the end of the AD experiments carried out with or without μAIR, respectively. As reported in Fig. 5a, the increase in biogas production due to the μAIR can be ascribed to the improved bioconversion of the VS contained in the inoculum, rather than to the presence of the PLA. In fact, compared to the CTRL, no significant changes in biogas production were observed in the presence of P-PLA, and only a slight increase of the biogas production was observed with UnP-PLA filaments (Fig. 5a). Accounting for the AD experiments carried out in the presence of UnP-PLA, the analysis of the biogas highlighted a significant increase of the cumulative CO_2 production (Fig. 5b). Although μAIR-AD did not cause significant biodegradation of the polymer, it would appear to have had some effect on the chemical features of the surface of the UnP-PLA filaments. To validate this hypothesis, IR analysis of the PLA samples after μAIR-AD was carried out.

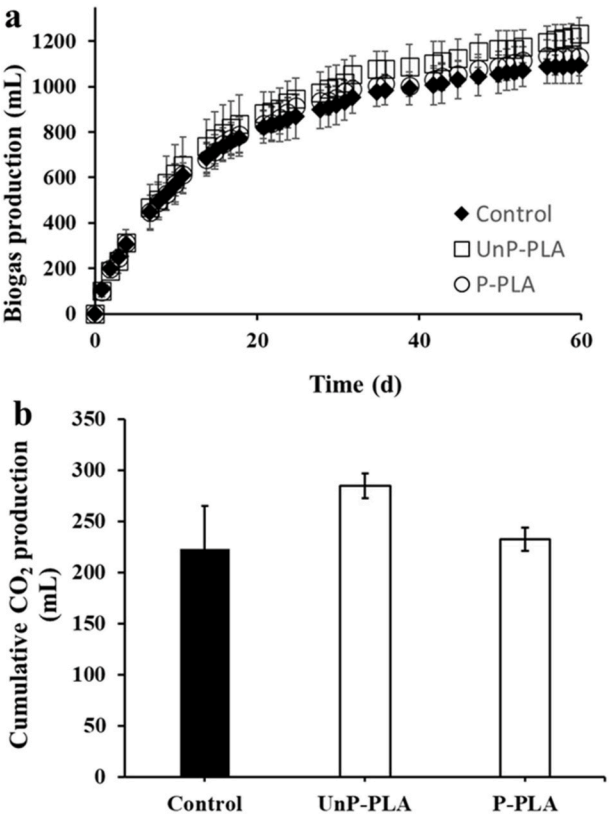


Fig. 5. Effect of micro-aeration on the AD of PLA: biogas production (a) and cumulative CO_2 production (b).

The IR spectra for UnP-PLA and P-PLA after μAIR-AD (Fig. S3 of the Supplementary information) showed profiles that were largely superimposable. No variation could be observed at the ν(C=O) band, which remained unaltered, with similar intensity and centered at ~1747 cm^{-1}. The mixed C-O-C/C-C-O/CH_3-coupled region at ~1210-1040 cm^{-1} and the crystalline sensitive region (980-900 cm^{-1}) maintained almost the same area ratio (IR-C_{index} remained within experimental error, for 3D-printed samples: anaerobic 4.2, micro-aerated 4.4; for unprinted: anaerobic 6.8, micro-aerated 6.4, see Table S2 of the Supplementary information). In addition, thermograms of PLA subjected to μAIR-AD (not shown) were almost analogous to those of PLA after a conventional AD, either P-PLA or UnP-PLA, with slightly lower crystallization and melting temperature than pristine PLA, as documented by data reported in Table 4.

Consequently, the higher CO_2 production observed during μAIR-AD was therefore best ascribed to process-level effects (oxygen-stimulated hydrolytic/acidogenic activity, re-oxidation of reduced inhibitors, and a more stable redox environment) rather than to any additional conversion of PLA. This is consistent with findings in the literature [63–65], where it was shown that μAIR enhances hydrolysis and acidogenesis, suppresses accumulation of inhibitory reduced species, and stabilizes oxidation-reduction potential in AD systems. Additionally, the limited oxygen dissolution, as facultative bacteria rapidly consumed the solubilized O_2, likely contributed to microbial aggregate buildup, as well as the tolerance of some methanogens to oxygen-inhibiting effects, thereby improving biogas generation [63].

The limited beneficial effect of micro-aeration can be attributed to the intrinsic structural properties of PLA and its low susceptibility to oxidative and enzymatic attack under anaerobic conditions. Indeed, micro-aeration was successfully applied to enhance AD of other recalcitrant substrates, such as LM. Facultative hydrolytic microorganisms, extracellular enzyme production, electron-transfer processes, and humic substance formation were probably less adequate for synthetic hydrophobic polyesters such as PLA [35]. Indeed, LMs contain naturally biodegradable polysaccharides that can be accessed by microorganisms under improved conditions, whereas PLA degradation remains strongly limited by slow ester bond hydrolysis and limited microbial accessibility [66]. Also, the applied air dosage of approximately 120 mL air/g VS led to oxygen dosages comparable to 0.38 mL O_2/g VS/day reported as effective for LM [35]. Therefore, the limited PLA conversion obtained in this work cannot be ascribed to an insufficient oxygen supply during micro-aeration operation. Further oxygen supply could have instead been negative on biogas production, as reported by Zhu et al., when exceeding 1.6 mL/g VS/day [18], which is not the case since biogas yield per g SV in the control was 0.5 times higher compared to control without micro-aeration strategy.

3.4. Effect of Fenton pretreatment

The Fenton reaction is usually used in AD to degrade recalcitrant organic compounds [67] and to enhance the methane production from slow biodegradable substrates [68,69]. The Fenton pretreatment was tested to break polymeric chains and improve PLA biodegradability. At the end of the Fenton pretreatment, both UnP-PLA and P-PLA did not show a significant decrease of their weight (<1%). In addition, the FT-IR spectra of both UnP-PLA and P-PLA samples after the Fenton pretreatment (Fig. S4 of the Supplementary information) did not show any alteration compared to the pristine materials. However, in the presence

of PLA, a significant enhancement of the soluble COD was observed within Fenton solutions. In fact, COD concentrations were 39.4 ± 2 mg/L, 117.9 ± 6.2 mg/L, and 91.9 ± 5.9 mg/L in CTRL, UnP-PLA, and P-PLA solutions, respectively. Although there was no weight loss of the PLA sticks after the AD, the higher biogas production was observed in the presence of Fenton pretreated UnP-PLA filaments (Fig. 6a). Also, the average CO_2 production increase was generally detected in the presence of PLA. In detail, an 8.7% and 11.9% increase in cumulative CO_2 production was observed compared to the CTRL in the presence of Fenton pretreated UnP-PLA and P-PLA, respectively (Fig. 6b). In addition, the cumulative CH_4 production during the AD treatment of Fenton pretreated P-PLA was higher than that of the CTRL (Fig. 6c).

This phenomenon could be ascribed to the presence of Fe(III) species, by-products of the Fenton treatment, which can oxidize recalcitrant organic compounds, such as PLA, through the intervention of iron-reduction bacteria [70]. However, the slight increases in biogas production during the AD of Fenton pretreated UnP-PLA and P-PLA highlighted only a negligible oxidation of the polymer surface. In fact, the DSC analysis on samples of Fenton pretreated PLA at the end of AD (Table 4 and Tables S1–S2 of the Supplementary information) did not show any significant changes compared to the untreated samples.

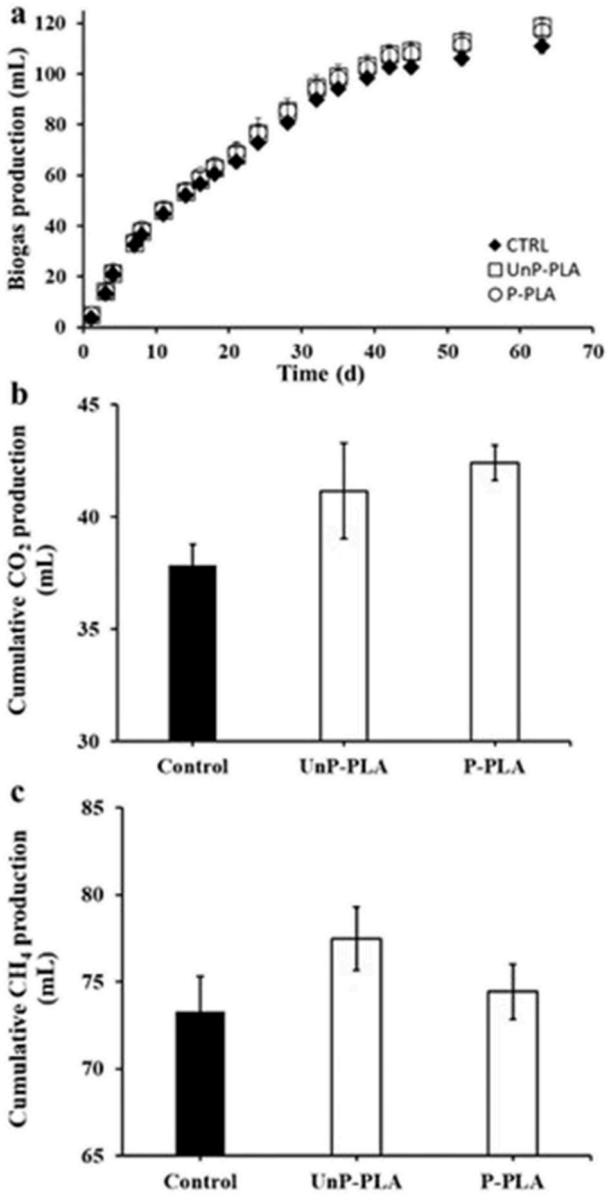


Fig. 6. Effect of Fenton pretreatment on the AD of PLA: biogas production (a), cumulative CO_2 (b) and CH_4 (c) productions.

The limited action of the Fenton pretreatment here achieved would indicate that the generated hydroxyl radicals mostly induced superficial oxidation phenomena without promoting extensive bulk depolymerization of the PLA matrix. Although the increase of soluble COD, a slight improvement of biogas generation was reached (Fig. 6). Based on the measured COD increase and the theoretical conversion factor of 0.35 mL CH_4/mg COD, the observed improvement is within the expected range, given the experimental variability of batch tests [71]. The irrelevant weight loss and the absence of substantial modifications in FT-IR (Fig. S4 of the Supplementary information) and DSC analyses can further suggest that the polymer backbone remained largely preserved after treatment [72]. This behavior can be attributed to the intrinsic physical-chemical characteristics of PLA, which is relatively chemically inert towards water and organic solvents, with limited accessibility of the polymer bulk to reactive species [73]. Also, heterogeneous Fenton approaches based on 3D-printed PLA structures showed that Fe-based catalytic sites can be mostly distributed on the external scaffold surface rather than homogeneously throughout the polymer matrix [74]. Thus, hydroxyl radical generation was expected to likely occur mainly at the polymer-liquid interface, limiting radical penetration into the inner PLA structure.

3.5. Effect of thermo-alkaline pretreatment

Table 6 shows the results in terms of weight loss and solubilization percentage of the $Ca(OH)_2$ thermo-alkaline (AT) pretreatment tests carried out on PLA at varying operating temperatures and pretreatment durations. Specifically, the solubilization percentage was evaluated as the ratio between the soluble COD concentration at the end of the tests ($[COD_l]$) and the COD of the PLA at the beginning of the experiments ($[PLA_0]$) (Eq. (5)):

$$Solubilization\ percentage = \frac{[COD_l]}{[PLA_0] \cdot 1.5\ \frac{g_{COD}}{g_{PLA}}} \times 100 \quad (5)$$

In agreement with other studies [27,28], as the temperature and treatment duration increased, both the weight loss and the solubilization percentage of PLA increased (Table 6). The solubilization of UnP-PLA at 70 °C for 48 h was 17.6%, which is a value lower than that reported by Cazaudehore et al. [28] (i.e., 49 ± 6 %). This discrepancy likely reflects differences in sample shape and size. In the present study, PLA sticks were used rather than granules, resulting in larger dimensions and a lower specific surface area, which represents an important parameter for hydrolysis kinetics [75].

$Ca(OH)_2$ AT pretreatment hydrolyzed PLA polymer with a consequent solubilization of the organic matter. Accordingly, solubilization percentages close to weight-loss percentages were observed at the end of each test. It is noteworthy that P-PLA was more affected by the AT treatment than UnP-PLA. In fact, higher P-PLA weight losses and solubilization percentages were always observed for all AT pretreatment conditions. The substantial effect obtained for P-PLA after thermo-alkaline pretreatment can be attributed primarily to the FDM-induced thermomechanical chain scission process, which likely reduced the average molecular weight by approximately 30% and improved the susceptibility of ester bonds to alkaline hydrolysis [76], rather than to the slight increase in specific surface area, which may have contributed only marginally to the accessibility of OH^- ions [77].

Specifically, the 72 h lasting pretreatment at 70 °C resulted in weight losses of 32.8% and 52.1%, and solubilization percentages of 22.5% and 39.1% for UnP-PLA and P-PLA sticks, respectively (Table 6).

As shown in Fig. 3, a macroscopic difference can be noticed in the thermograms of PLA samples after AT. Only the glass transition at 61.3 and 63 °C and melting at 156.7 and 155.3 °C for the P-PLA and UnP-PLA

Table 6
Thermo-alkaline pretreatment of UnP-PLA and P-PLA.

35 °C						
Sample	24 h		48 h		72 h	
	Weight loss (%)	Solubilization percentage (%)	Weight loss (%)	Solubilization percentage (%)	Weight loss (%)	Solubilization percentage (%)
UnP-PLA	3.1 ± 0.3	2.8 ± 0.2	8.1 ± 0.3	5.4 ± 0.5	11.2 ± 0.5	9.3 ± 0.7
P-PLA	5.6 ± 0.2	2.9 ± 0.3	9.5 ± 0.4	6.2 ± 0.4	13.1 ± 0.5	9.8 ± 0.6
70 °C						
Sample	24 h		48 h		72 h	
	Weight loss (%)	Solubilization percentage (%)	Weight loss (%)	Solubilization percentage (%)	Weight loss (%)	Solubilization percentage (%)
UnP-PLA	12.3 ± 0.5	10.4 ± 0.5	23.3 ± 0.7	17.6 ± 0.5	32.8 ± 1.3	22.5 ± 0.7
P-PLA	16.0 ± 0.6	12.3 ± 0.6	32.2 ± 0.9	22.6 ± 0.6	52.1 ± 1.7	39.1 ± 1.1

samples, respectively, were observed. In fact, the aging and crystallization peaks disappeared due to the AT treatment operated at 70 °C. This finding clearly evidences that the AT treatment induced crystallization of the amorphous phase, as highlighted by the intense melting peaks, corresponding to a high crystallinity content of 44.7 and 50.0% (Table 4). Consequently, the melting temperature increased with respect to the previous samples. Moreover, the second DSC scan of P-PLA and UnP-PLA (after AT), highlighted (i) the crystallization peak T_c decrease from about 116 to 119 °C of the previous PLAs to 107.2 and 111.3 °C, suggesting a decrease of molecular weight after AT, and (ii) the double melting peak at about 147 and 155 °C, ascribable to domains with different molecular weight (see Fig. S5 of the Supplementary information). Specifically, after AT, a relevant reduction of the molecular weight to 14140 and 9350 g/mol (intrinsic viscosity decreased to about 0.4-0.6 dL/g) could be observed (Table 5), in agreement with the DSC results.

The FT-IR spectra (Fig. 7) of PLA samples subjected to AT showed significant structural modifications, in agreement with DSC and solution viscometry results.

A single carbonyl absorption band centered at ~1747 cm^{-1} changed upon AT into a clear double and overlapping bands with maxima at ~1755 and ~1747 cm^{-1}. The formation of ν(C=O) centered at 1755 cm^{-1} suggests (i) that ester carbonyls were in more ordered environments (i.e., a more crystalline/packed configuration), and/or (ii) the presence of new functional groups such as -COOH, derived from hydrolysis. The band centered at ~1267 cm^{-1} ($\nu(C–O)_{ester}+\delta(CH)$, more intense in amorphous configurations) diminished, and the shoulder at ~1208 cm^{-1} (ν_{as}(C-O-C)) became more evident, reflecting a redistribution among backbone modes as chains pack more tightly. The IR spectra of the AT sample also showed new bands centered at ~921 and ~698 cm^{-1} and an increase of A_{1126}/A_{1041} within the mixed C-O-C/C-C-O/CH_3-coupled manifold. The increase in the A_{1126}/A_{1041} ratio reflects greater conformational order ($r_{as}(CH_3)/\nu$(C-CH_3), while the ~921 cm^{-1} formation, with the band centered at ~956 cm^{-1} decreasing, implies significant growth of the 10_3 helical crystalline population (IR-C_{index} upon AT increases to 15.5 and 24.2 for UnP-PLA and P-PLA, respectively) (Table S2 of the Supplementary information). Then, the peak at ~698 cm^{-1} is ascribed to carbonyl out-of-plane bending mode (γ(C=O)) that sharpens as crystallinity increases [58–60].

Results on mesophilic AD of PLA sticks pretreated at 70 °C for 72 h (AT) are reported in Fig. 8. In this case, a great enhancement of the biogas production was observed for both pretreated UnP-PLA and P-PLA sticks. In detail, the AD of AT pretreated UnP-PLA resulted in a net CO_2 and CH_4 production of 65 mL (100 mL/g_{PLA}) and 208 mL (319 mL/g_{PLA}), respectively. The CO_2 and CH_4 net productions further increased in the presence of pretreated P-PLA by 42% and 13%, respectively. This enhanced biomethane production was consistent with the structural modifications observed by FT-IR, DSC, and viscometry analyses, which

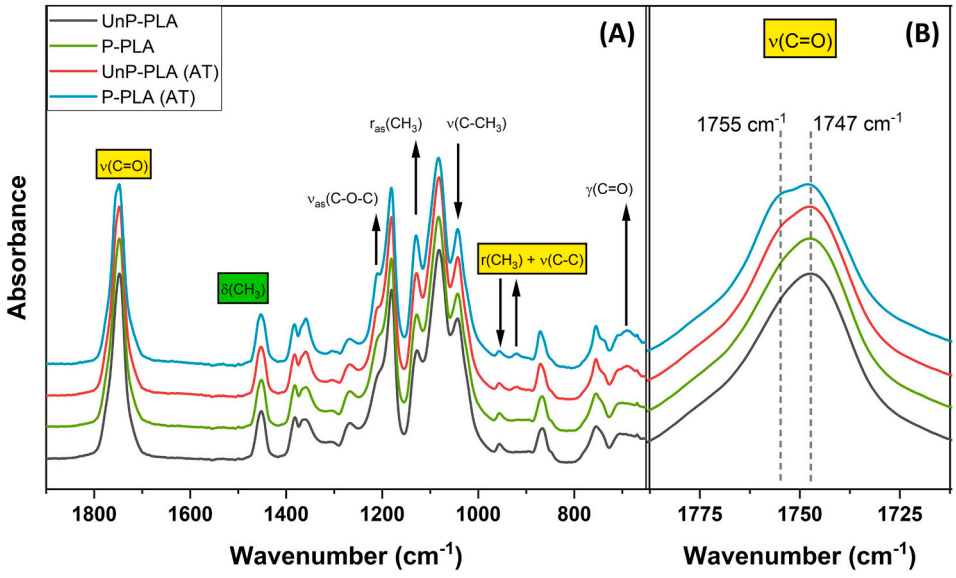


Fig. 7. Panel A: FT-IR spectra (1900-650 cm^{-1}) normalized to the ~1454 cm^{-1} band ($\delta(CH_3)$, highlighted in green) for unprinted (UnP-PLA, black) and printed (P-PLA, green) PLA before and after alkaline pretreatment (UnP-PLA (AT) red, P-PLA (AT) cyan). Key normal modes with the largest changes are highlighted (yellow). Panel B: insert showing carbonyl-related bands in detail. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

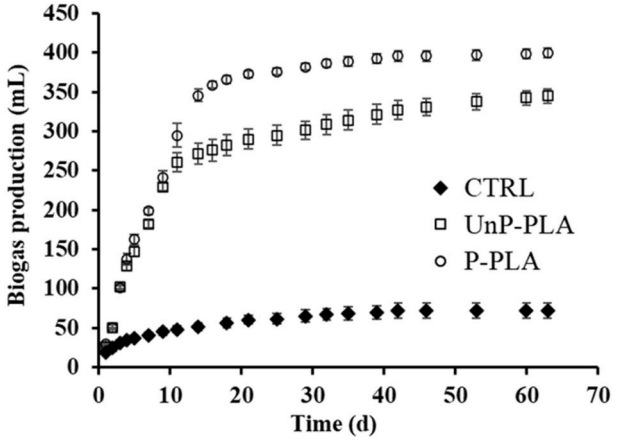


Fig. 8. Effect of $Ca(OH)_2$ thermo-alkaline pretreatment on the biogas production during the AD of PLA.

suggested ester-bond cleavage, molecular-weight reduction, and increased PLA depolymerization after thermo-alkaline pretreatment [78].

Although during the first days there were no significant differences in biogas production between the AD of pretreated UnP-PLA and P-PLA, from the 10th day, biogas production for UnP-PLA started to slow down (Fig. 8), probably due to the conversion of the whole soluble COD amount. Instead, in the case of pretreated P-PLA, characterized by a higher solubilization percentage, the biogas production slowed down only from the 14th day.

After the AD, the PLA sticks appeared very brittle and fragmented (Fig. 9). There was a further weight loss compared to the initial weight of PLA sticks, equal to 33.5% and 31% for UnP-PLA and P-PLA, respectively. Thus, a whole weight loss of 66.3 ± 1.3% and 83.1 ± 1.3% was observed for UnP-PLA and P-PLA, respectively.

The above-reported results agree with another study [28] where the AT pretreatment on UnP-PLA granules (2.5% w/v $Ca(OH)_2$, 70 °C, 48 h) resulted in a solubilization percentage of 73% in 30 d and a net biomethane production of 367 mL/g_{PLA}. A higher solubilization of UnP-PLA (>90%) was obtained after an alkaline pretreatment lasting 7 d and carried out with 1 M NaOH and 37 °C, but the subsequent mesophilic AD resulted in a similar biomethane yield (360-380 mL/g_{PLA}) [11].

Upon both AT and AD, X_C^{1st} increased to 64.5% and 52.6% for UnP-PLA and P-PLA samples, respectively, while T_m decreased to the lowest value of about 144 °C for both the samples (Table 4), consistent with lamellar thinning after extensive chain scission [79]. This result suggests that further molecular weight reduction was now prevailing over crystallinity in affecting the melting temperature decrease. Accordingly, in the second DSC scan (Fig. S5 and Table S1 of the Supplementary information), a relevant decrease of crystallization peak to 103.5-108.2 °C was observed.

Notably, the Δc_p values listed in Table 4 closely reflect the observed crystallinity trends, being lower for the samples with higher crystallinity (AT and AT + AD). In fact, they are related to the intensity of the second-order phase transition, i.e., to the content of the amorphous phase. At the same time, the highest values of crystallizability of 32-44% were also found for samples AT and AT + AD, consistent with a decrease of molecular weight.

Glass transition temperature was found to be dependent on the treatment, with values of 67-69 °C for AD, about 66 °C for μAIR-AD, and about 62 °C for AT samples. The double treatment AT + AD determined ever lower values of T_g, i.e., 56.5 and 52.8 °C for UnP-PLA and P-PLA, respectively. Moreover, a further decrease of T_g was also observed in the cooling step and in the second DSC scan. This trend specifically reflects the behavior of an almost fully amorphous PLA and is more informative of the modifications of the molecular weight of the PLA chains, either in the molten state after melting during cooling, or in the glassy state during the second heating. These findings are confirmed by the viscometry data reported in Table 5. It is worth noting that even lower intrinsic viscosity values could be estimated for the samples treated with AT + AD, roughly corresponding to molecular weight values far below 3000 g/mol. Therefore, being below the limit of validity of the used approach, these values are not reported in Table 5, but they are indicative of a strong reduction of the molecular weight after AT + AD treatment, in agreement with the DSC results. These structural modifications likely improved water penetration and microbial accessibility to the polymer matrix, thus facilitating anaerobic hydrolysis and methane generation [80].

Taken together, the consistently low crystallinity degree and the modest molecular weight change indicate that both printing and mesophilic AD, under the conditions involved, did not result in measurable conversion or bulk reorganization of PLA. On the contrary, samples subjected to AT and, to a greater extent, to AT + AD showed a significant reduction of the average molecular weight.

Consistent with macroscopic evidence, the IR spectra of the samples subjected to AT + AD show that significant changes occurred in the physical-chemical properties of PLA (Fig. 10).

As in the AT samples, the carbonyl region presented two overlapping bands, centered at 1755/1747 cm^{-1}. Moreover, a weak broad band appeared at ~1636 cm^{-1}, consistent with δ(H-O-H) of adsorbed water; its presence is consistent with increased polarity after hydrolysis and enhanced moisture uptake by hydrolysis-derived end-groups. A growth at ~1360 cm^{-1}, assigned to $\delta_S(CH_3)$ coupled with backbone deformations that intensify as chains pack, accompanied by the loss of the

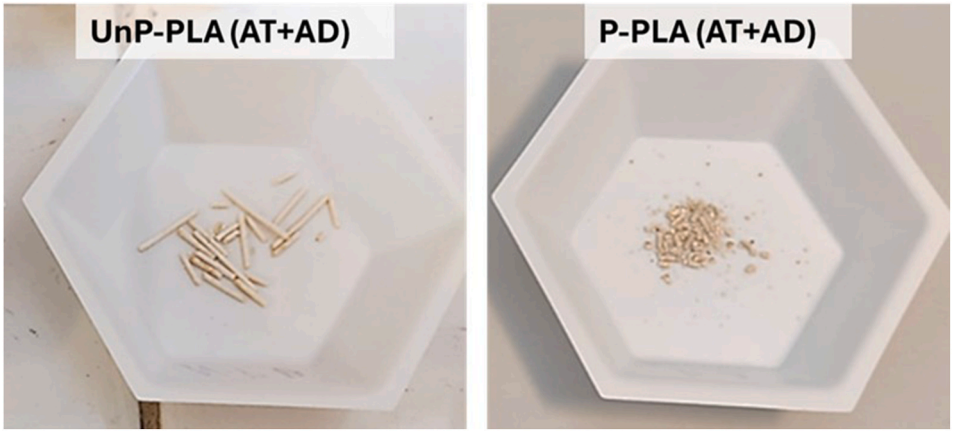


Fig. 9. UnP-PLA and P-PLA sticks after the thermal-alkaline pretreatment at 70 °C for 72 h and mesophilic AD lasting 63 d.

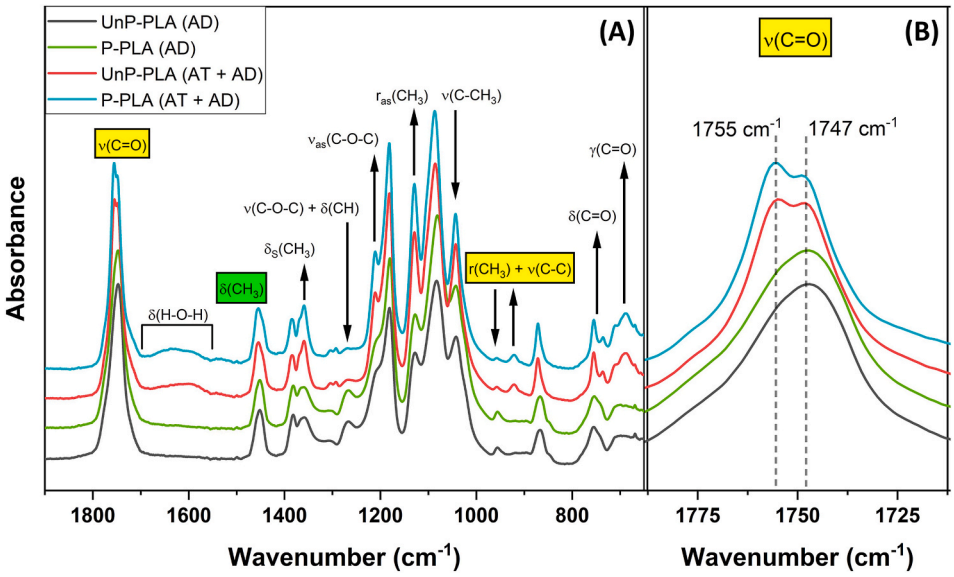


Fig. 10. Panel A: FT-IR spectra (1900-650 cm^{-1}) of PLA digested (AD) samples, normalized to the ~1454 cm^{-1} band ($\delta(CH_3)$, highlighted in green): unprinted (UnP-PLA (AD), black) and printed (P-PLA (AD), green) PLA; after alkaline pretreatment (UnP-PLA (AT + AD) red, P-PLA (AT + AD) cyan). Key normal modes with the largest changes are highlighted (yellow). Panel B: insert showing carbonyl related bands in detail. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

band centered at ~1267 cm^{-1} (ν(C-O-C) + δ(CH)) implies, also in this case, a shift from disordered chain configuration to a more organized one. In parallel, the shoulder visible at ~1210 cm^{-1} was sharper, and the A_{1126}/A_{1041} ratio increased, reflecting greater conformational order. Moreover, in the crystallinity window, the 956 cm^{-1} component diminished while the one at 921 cm^{-1} grew, directly tracking the amorphous-to-crystalline redistribution (IR-C_{index} upon AT and AD increases to 47.0% and 55.8% for UnP-PLA and P-PLA, respectively) (Table S2 of the Supplementary information). In parallel, the ~754/734 cm^{-1} doublet (Fig. 10), assigned to carbonyl in-plane bending (δ(C=O)) coupled to lattice/helix deformations, sharpened with increasing crystallinity, complementing the γ(C=O) out-of-plane bend at ~698 cm^{-1} and evidencing lamellar/crystal organization and longer-range order [58–60]. These infrared spectroscopic results align quantitatively with DSC and viscometry investigations. In fact, the visible spectral changes and the thermal/viscometric features support a coherent mechanism: alkaline scission produces shorter chains that chemi-crystallize, and subsequent AD does not reverse this ordering but rather consolidates existing crystallites [79], while the smallest and soluble fragments are likely transferred and bio-processed in the liquid medium. The effect is systematically stronger for P-PLA (processed at 195 °C), ascribable to its printing-induced defects that facilitate ester-bond hydrolytic cleavage.

As shown in Table 4, the crystallinity enhancement, particularly in AT + AD samples, can be ascribed to the preferential degradation of amorphous regions of the polymer. The amorphous fractions are weaker, structurally less ordered, and more susceptible to degradation by chemical and biological agents. Therefore, the more rapid removal of these parts leads to a relative increase in the percentage of residual crystalline parts [81,82].

3.6. Kinetic interpretation of methane production

To further investigate the effects of the different process-based strategies on methanation dynamics and microbial adaptation phenomena, the cumulative methane yield profiles were fitted using the modified Gompertz equation (Eq. (6)), which is widely applied for AD kinetics [83]:

$$G(t) = a \cdot \exp\left[-\exp(-k(t - x_c))\right] \quad (6)$$

where "$G(t)$" is the cumulative methane production at time "t" (day), "a" represents the ultimate methane potential (mL), "k" is the methane production rate (1/day), and "xc" is the acclimation time (days).

The results (Table 7) suggested that in all the investigated conditions, the modified Gompertz model is suitable for describing AD dynamics due to the high adjusted R^2 values (i.e., 0.975-0.996).

In untreated samples, only limited kinetic differences were observed among the CTRL, UnP-PLA, and P-PLA conditions. In particular, P-PLA showed a slightly higher methane potential (i.e., a = 73.75 mL) and kinetic constant (i.e., k = 0.106 1/day) than UnP-PLA and CTRL, together with a slightly lower lag-phase value (i.e., x_c = 7.20 days). Also, the modified Gompertz modeling confirmed the limited effectiveness of micro-aeration for PLA biodegradation. Indeed, the kinetic constant remained almost unchanged (i.e., k = 0.105-0.118 1/day). Similarly, the Fenton pretreatment did not significantly affect methane production kinetics, as comparable "a", "k", and "x_c" values were obtained for CTRL, UnP-PLA, and P-PLA samples (Table 7). On the contrary, thermo-alkaline pretreatment significantly affected methane production kinetics. In particular, the highest methane-production potential was achieved with P-PLA (i.e., a = 265.70 mL), whereas the kinetic constant was improved from 0.095 to 0.209 1/day in CTRL and P-PLA, respectively (Table 7).

Therefore, methane production kinetics increased by more than 2-fold after thermo-alkaline pretreatment (Table 7). The kinetically relevant stages for AD are hydrolysis, acidogenesis, acetogenesis, and methanogenesis, which are coupled with specific, functionally specialized microbial strains [84]. Hydrolytic and fermentative bacteria (e.g., *Clostridium*, *Bacteroides*, *Ruminococcus*, *Fibrobacter*, *Lactobacillus*, and *Enterobacter*) probably dominated hydrolysis and acidogenesis [85]. Accordingly, the improved methane-production kinetics (Table 7) likely resulted from enhanced hydrolysis and increased microbial accessibility to the PLA matrix (Fig. 10). Afterwards, syntrophic acetogenic bacteria (e.g., *Syntrophomonas* and *Syntrophobacter*) were likely to associate acetogenesis with H_2 removal, and methanogenic archaea (e.g., *Methanosaeta* and *Methanosarcina*) can subsequently conduct the final conversion of reaction metabolites to biogas [85]. The lower "x_c" values obtained for thermo-alkaline-treated PLA samples (Table 7) suggested a faster transition toward active methanogenesis.

Table 7
Kinetic parameters obtained from the modified Gompertz modeling of cumulative methane yield during the AD of control (CTRL), unprinted PLA (UnP-PLA), and printed PLA (P-PLA) under conventional AD, and process-based strategy conditions.

Process-based strategy	Experimental conditions	Gompertz parameters			Adj. R^2
		a [mL]	x_c [day]	k [1/day]	
Untreated	CTRL	69.68 ± 2.10	8.06 ± 0.69	0.0899 ± 0.0090	0.9765
	UnP-PLA	69.65 ± 1.93	7.97 ± 0.62	0.0942 ± 0.0088	0.9793
	P-PLA	73.75 ± 1.92	7.20 ± 0.57	0.1058 ± 0.0100	0.9787
Micro-aeration	CTRL	867.48 ± 8.51	7.84 ± 0.28	0.1082 ± 0.0049	0.9869
	UnP-PLA	935.61 ± 11.24	6.36 ± 0.35	0.1184 ± 0.0073	0.9749
	P-PLA	910.81 ± 11.73	7.00 ± 0.37	0.1052 ± 0.0064	0.9763
Fenton oxidation	CTRL	75.33 ± 1.60	15.30 ± 0.48	0.0718 ± 0.0037	0.9947
	UnP-PLA	79.78 ± 1.61	16.00 ± 0.46	0.0700 ± 0.0033	0.9957
	P-PLA	76.16 ± 1.36	15.85 ± 0.40	0.0727 ± 0.0032	0.9963
Thermo-alkaline	CTRL	37.33 ± 0.41	6.69 ± 0.26	0.0952 ± 0.0042	0.9941
	UnP-PLA	225.07 ± 3.92	5.03 ± 0.33	0.1919 ± 0.0188	0.9728
	P-PLA	265.70 ± 2.02	6.05 ± 0.14	0.2090 ± 0.0088	0.9954

3.7. Potential industrial application, limitations, and challenges

A preliminary energy balance was performed to support the potential industrial application of this approach by estimating whether the energy theoretically available from methane produced during AD can account for the energy required by the process-based strategy employed here (i.e., thermo-alkaline pretreatment).

The thermal energy required to heat the alkaline pretreatment medium from room temperature to 70 °C was calculated through Eq. (7) [86]:

$$Q = m \cdot c_p \cdot \Delta T \quad (7)$$

where (Q) is the thermal energy required, (m) is the mass of the liquid phase (i.e., 13 g) or PLA (i.e., 0.65 g), (c_p) is the specific heat capacity of the water (i.e., 4.18 kJ/kg °C) or the PLA (i.e., 1.8 kJ/kg °C [87]), and (ΔT) represents the increase of temperature from 25 to 70 °C.

Therefore, the total theoretical heating demand was 2.5 kJ. Taking into account the biomethane produced from P-PLA after thermo-alkaline pretreatment and by subtracting the contribution of the control test during AD (i.e., approximately 271.77 and 36.66 mL, respectively), the net energy recoverable from P-PLA was estimated, assuming a CH_4 lower heating value (LHV) of 35.8 MJ/m^3 [88], as 8.42 kJ ($E_{CH,4}$).

Finally, a positive net energy (E_{net}) of 5.92 kJ is obtained by subtracting the energy needed for the process-based strategy (Q) from $E_{CH,4}$. This preliminary calculation suggests that methane produced from thermo-alkaline-pretreated P-PLA could offset the energy required for the pretreatment.

Thermo-alkaline strategies were previously associated with a positive energy balance during AD processes of other substrates. For instance, Kim et al. [22] reported that thermo-alkaline hydrolysis allowed for an energy gain of 161.4 ± 39.3 MJ/ton. Similarly, Budych-Gorzna et al. [89] reported an improvement of methane yield from 210.4 to 267.1 mL CH_4/gVS after low-temperature thermo-alkaline pretreatment of waste activated sludge, along with enhanced electricity recovery.

However, this calculation was intended only as a preliminary estimation and did not include heat losses, mixing energy, neutralization requirements, reactor thermal inertia, or scale-up effects [90]. On the other hand, at a larger scale, the E_{net} could be improved by heat recovery (i.e., 85%) or combined heat and power (CHP) systems using the produced biogas [91,92]. Assuming a CHP electrical conversion efficiency of 50% and an electricity value of 86 €/MWh [89], the E_{net} achieved with the further biomethane generated after thermo-alkaline pretreatment, equivalent to an estimated electrical value of about 1.8 kWh/kgPLA, could theoretically correspond to a revenue of approximately 0.16 €/kgPLA.

Therefore, these preliminary results indicate that a thermo-alkaline pretreatment here employed as a process-based strategy may have potential engineering feasibility if properly optimized and integrated into existing AD systems. However, some limitations may arise because, in the BMP test, important operating parameters, such as the organic loading rate (OLR) and hydraulic retention time (HRT), are unavailable in a batch system [93]. Indeed, depending on OLR, HRT, digester type, and other factors, the biogas produced in continuous approaches can differ from that produced in BMP systems. Besides volatile fatty acid accumulation, the washout effect and biomass accumulation, which affect the process stability of the AD, are not considered here. Therefore, further investigations under continuous or semi-continuous operating conditions are necessary to better evaluate the industrial applicability and operational feasibility of the proposed thermo-alkaline strategy.

From an environmental perspective, the global warming potential of AD depends on the energy vector used. In the case of biogas, after CO_2 removal using appropriate activated carbon filters [94], biomethane transportation can be coupled with a lower carbon footprint than heat or electricity production [95]. This is most likely because heat and electricity are considered lower-energy vectors, whereas the transportation sector relies more on fossil fuels [96].

Also, the thermo-alkaline process-based strategy requires chemical inputs and elevated temperatures, which can lead to an increase in operating complexity and parasitic energy demand [85], particularly at the laboratory scale, thereby increasing the environmental burden due to alkaline streams. In this sense, policy support can play a key role in making AD technologies more environmentally feasible with proper incentives. For instance, future investigations should also focus on enzymatic approaches as sustainable alternatives for improving the anaerobic conversion of 3D-printing-derived PLA.

4. Conclusions

This work investigated the effectiveness of different approaches for 3D-printing-derived waste biodegradability enhancement during the AD process. The aim of this study was to provide a feasible solution for the management of polymeric residuals, which are nowadays produced in significant amounts after the application of widely used 3D printing procedures. Among the different potential wastes, attention was focused on PLA as one of the main materials involved in 3D printing. Accounting for the effect of the 3D printing, although results showed a breakage of the printed polymer chains resulting in a 30% decrease of its average molecular weight, no remarkable differences were observed during AD of P-PLA compared to the UnP-PLA filaments. This further highlighted the significance of an investigation into proper process-based strategy of these wastes that, despite their physical-chemical alteration after a 3D printing phase, can still display recalcitrant features during biological conversion processes. According to the outcomes from this study, similar results after AD with μAIR or AD following a Fenton oxidation were observable on both P-PLA and UnP-PLA samples, thus suggesting a not

relevant effect of these strategies for the polymer bioconversion enhancement. On the contrary, encouraging results in terms of PLA biological conversion were observed by testing thermo-alkaline pretreatment before AD. In fact, after the AT treatment (2.5% $Ca(OH)_2$) of PLA at 70 °C for 72 h, UnP-PLA showed a 32.8% weight loss and a 22.5% solubilization. The same process-based strategy was even more efficient on P-PLA sticks, since a 52.1% weight loss and a 39.1% solubilization were obtained. Also, for these tests, FT-IR spectra, DSC analysis, and viscometry measurements highlighted remarkable structural modifications such as an increase in crystallinity and a decrease in the average molecular weight (>70%). This resulted in a further weight loss during the following AD treatment and in a biomethane production of 319 mL/g_{PLA} and 360 mL/g_{PLA} for UnP-PLA filaments and P-PLA sticks, respectively. In particular, after 63 d of AD on the AT pretreated UnP-PLA filaments and P-PLA sticks an overall weight loss of 66.3% and 83.1% were obtained, respectively. Based on the overall outcomes, this work outlined different levels of effectiveness displayed by the investigated treatment strategies for the enhancement of 3D-printing-derived waste bioconversion. However, the AD performance positive results, especially observed on the AT pretreated PLA, still suggest the viability of the biological process as a strategy for the valorization of this waste category. To strengthen the suitability of this approach, future works should focus on optimizing the 3D-printing process, evaluating how controlled modifications of the printing parameters could further enhance the effectiveness of pretreatments and improve the subsequent biodegradation of the polymeric waste. Besides this, the optimization of operating conditions is also fundamental for the potential valorization of different strategies. For this aim, the assessment of the process-based strategy effects on the polymeric waste is crucial. This indeed highlights, as also observed from the results of the present work, that an experimental approach integrated with material characterization analyses is of significant importance to carry out more comprehensive studies in this research field.

CRediT authorship contribution statement

Davide Giandomenico: Investigation, Writing – original draft. **Alessandro Gottuso:** Data curation, Formal analysis. **Rocco Rubino:** Resources, Validation. **Alberto Ferraro:** Methodology, Writing – review & editing. **Luca Fambri:** Formal analysis, Validation. **Francesco Parrino:** Conceptualization, Writing – original draft. **Raffaele Morello:** Data curation, Investigation. **Francesco Bianco:** Formal analysis, Methodology. **Gennaro Trancone:** Resources, Visualization. **Umberto Fratino:** Funding acquisition, Supervision. **Danilo Spasiano:** Conceptualization, Writing – review & editing.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biombioe.2026.109636.

Data availability

Data will be made available on request.

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